Perancangan Aplikasi Web Joki Genshin Impact

Dokumen analisis & desain sistem: requirement, ERD, DFD, DDL, tech stack, sitemap, dan user flow.

0. Keputusan Desain & Saran (yang Disederhanakan)

Sesuai permintaan, bagian yang kurang penting tidak dibuat. Berikut keputusannya:

Usulan awal	Keputusan	Alasan
Tabel `admins` + `roles`	**Dihapus**, diganti kolom `users.role` (`user`, `admin`, `booster`)	Hanya 3 peran tetap, tidak butuh RBAC dinamis. Tabel terpisah menambah JOIN tanpa manfaat.
`order_items` (multi-item)	**Dihapus**, 1 order = 1 paket	Jasa joki biasanya dipesan satuan. Lebih sederhana untuk antrian, harga, dan review. Jika butuh banyak layanan, user membuat beberapa order.
Tabel `queue`	**Dihapus**, antrian dihitung dari `orders` (status `queued`, urut `queue_priority` lalu `paid_at`)	Tabel terpisah rawan tidak sinkron dengan status order.
Tabel `invoices`	**Dihapus**, invoice dibuat (PDF) dari `orders` + `payments`	Data invoice sudah ada. Nomor invoice = `order_no`.
Tabel `settings`	**Dihapus**, pakai file `.env`/config	Tidak ada pengaturan dinamis yang penting di MVP.
Tabel `leaderboard`	**Diganti VIEW** (opsional cache Redis)	Data turunan dari `orders`, menyimpannya = redundansi.
`activity_logs`	**Dipertahankan versi ringan**	Penting untuk audit refund, ubah status, dan ban.
Booster/talent	**Ditambah** `users.role = 'booster'` + `orders.booster_id`	Memungkinkan owner punya beberapa pengerja tanpa tabel baru.

Saran tambahan:

1. **Kredensial akun game** disimpan terenkripsi AES-256-GCM (kunci di env/KMS, bukan di DB), dalam tabel terpisah, dan **dihapus otomatis setelah order selesai** (+7 hari).
2. **Pakai payment gateway otomatis (Midtrans)** untuk MVP. Transfer manual dijadikan opsi cadangan karena butuh verifikasi manual admin.
3. **Harga disalin (snapshot)** ke order agar perubahan harga paket tidak mengubah riwayat/report keuangan.
4. **Leaderboard**: tambahkan opsi *tampilkan sebagai anonim* (`users.show_on_leaderboard`) demi privasi. Hitung hanya order berstatus `completed`, dikurangi refund.
5. **Risiko bisnis**: joki melanggar ToS HoYoverse (akun bisa di-ban). Tampilkan *disclaimer + persetujuan (checkbox)* saat checkout.
6. Fitur yang sebaiknya ditunda ke fase 2: review, filter leaderboard mingguan/bulanan, export Excel (PDF dulu), notifikasi email/push.

1. Requirement

1.1 Functional Requirements

ID	Modul	Kebutuhan	Prioritas
FR-01	Auth	Register, login, logout	Wajib
FR-02	Auth	Verifikasi email dan reset password via token	Wajib
FR-03	Profil	Edit nama, email, foto profil, ganti password	Wajib
FR-04	Katalog	Lihat kategori & paket joki (harga, durasi, deskripsi, tier)	Wajib
FR-05	Order	Pilih paket, isi data akun game (UID, server, kredensial terenkripsi), catatan, upload lampiran	Wajib
FR-06	Order	Checkout dan persetujuan disclaimer	Wajib
FR-07	Payment	Bayar via gateway (QRIS/VA/e-wallet) atau transfer manual + upload bukti	Wajib
FR-08	Payment	Status pembayaran: pending, paid, failed, refunded	Wajib
FR-09	Antrian	Lihat posisi antrian dan estimasi waktu tunggu	Wajib
FR-10	Order	Status: menunggu pembayaran, antri, dikerjakan, selesai, dibatalkan	Wajib

ID	Modul	Kebutuhan	Prioritas
FR-11	Chat	Chat real-time user ↔ admin, riwayat tersimpan, indikator belum dibaca	Wajib
FR-12	Leaderboard	Ranking berdasarkan total pengeluaran (rank, username, total, jumlah order)	Wajib
FR-13	Invoice	Lihat detail order dan unduh invoice PDF	Wajib
FR-14	Review	Rating & ulasan setelah order selesai	Opsional
FR-15	Admin	Dashboard: total order, revenue, order aktif, user baru	Wajib
FR-16	Admin	CRUD kategori dan paket joki	Wajib
FR-17	Admin	Kelola order: ubah status, assign booster, batalkan, refund	Wajib
FR-18	Admin	Atur urutan/prioritas antrian	Wajib
FR-19	Admin	Kelola user: lihat detail, ban/unban, reset password	Wajib
FR-20	Admin	Verifikasi pembayaran manual	Wajib
FR-21	Admin	Inbox chat semua user	Wajib
FR-22	Admin	Report keuangan mingguan: pendapatan, breakdown per jenis joki, sukses/gagal/refund, grafik, export PDF (Excel opsional)	Wajib
FR-23	Admin	Reset/atur periode leaderboard	Opsional
FR-24	Admin	Log aktivitas	Opsional

1.2 Non-Functional Requirements

ID	Kategori	Kebutuhan
NFR-01	Keamanan	Password di-hash **argon2id/bcrypt**; kredensial game dienkripsi AES-256-GCM
NFR-02	Keamanan	HTTPS/TLS wajib, HSTS, cookie `HttpOnly` + `Secure` + `SameSite`
NFR-03	Keamanan	Validasi input server-side, proteksi SQL injection & XSS, CSRF token

ID	Kategori	Kebutuhan
NFR-04	Keamanan	Rate limiting (login, reset password, chat, API), verifikasi signature webhook payment
NFR-05	Keamanan	Otorisasi berbasis peran; user hanya boleh akses order/chat miliknya
NFR-06	Performa	Response halaman/API p95 < 2 detik; index pada kolom filter/pencarian
NFR-07	Skalabilitas	Aplikasi stateless (session di Redis) agar bisa horizontal scaling; ≥ 500 concurrent user
NFR-08	Usability	Responsive (mobile-first), UI Bahasa Indonesia
NFR-09	Ketersediaan	Backup DB harian otomatis (retensi 14 hari) + uji restore berkala; uptime ≥ 99%
NFR-10	Maintainability	Migrasi DB terversi, logging terpusat, environment dev/staging/prod
NFR-11	Privasi	Data kredensial game dihapus otomatis setelah order selesai

2. ERD

```
erDiagram
    users ||--o{ orders : "memesan (user_id)"
    users |o--o{ orders : "mengerjakan (booster_id)"
    joki_categories ||--o{ joki_packages : "memiliki"
    joki_packages ||--o{ orders : "dipesan sebagai"
    orders ||--o| order_game_accounts : "menyimpan akun"
    orders ||--o{ payments : "dibayar lewat"
    orders ||--o| reviews : "diulas"
    users ||--o{ reviews : "menulis"
    users ||--o| conversations : "memiliki"
    conversations ||--o{ messages : "berisi"
    users ||--o{ messages : "mengirim"
    orders |o--o{ messages : "dirujuk"
    users |o--o{ activity_logs : "melakukan"

    users {
        bigint id PK
        varchar name
        varchar email UK
        varchar password_hash
```

```
        varchar avatar_url
        enum role "user|admin|booster"
        enum status "active|banned"
        boolean show_on_leaderboard
        timestamp email_verified_at
        timestamp created_at
    }
    joki_categories {
        bigint id PK
        varchar name UK
        varchar slug UK
        text description
        boolean is_active
    }
    joki_packages {
        bigint id PK
        bigint category_id FK
        varchar name
        text description
        numeric price
        int estimated_hours
        boolean is_active
    }
    orders {
        bigint id PK
        varchar order_no UK
        bigint user_id FK
        bigint package_id FK
        bigint booster_id FK "nullable"
        enum status
        numeric price_snapshot
        numeric total_amount
        int est_hours_snapshot
        int queue_priority
        text customer_notes
        timestamp paid_at
        timestamp started_at
        timestamp completed_at
    }
    order_game_accounts {
        bigint order_id PK, FK
        varchar uid
        enum server
        bytea credential_encrypted
        varchar attachment_url
        timestamp purge_at
```

```
    }
    payments {
        bigint id PK
        bigint order_id FK
        enum method "gateway|manual"
        varchar gateway_ref UK
        numeric amount
        enum status "pending|paid|failed|refunded"
        varchar proof_url
        numeric refunded_amount
        bigint verified_by FK "nullable"
        timestamp paid_at
    }
    reviews {
        bigint id PK
        bigint order_id FK, UK
        bigint user_id FK
        smallint rating
        text comment
    }
    conversations {
        bigint id PK
        bigint user_id FK, UK
        timestamp last_message_at
        int admin_unread
        int user_unread
    }
    messages {
        bigint id PK
        bigint conversation_id FK
        bigint sender_id FK
        bigint order_id FK "nullable"
        text body
        varchar attachment_url
        timestamp read_at
        timestamp created_at
    }
    activity_logs {
        bigint id PK
        bigint actor_id FK "nullable"
        varchar action
        varchar entity_type
        bigint entity_id
```

```
        jsonb meta
        inet ip
        timestamp created_at
    }
```

Leaderboard bukan tabel, melainkan VIEW `v_leaderboard_alltime` (lihat bagian 4).

2.1 Penjelasan Kardinalitas

Relasi	Kardinalitas	Keterangan
users → orders (pemesan)	1 : N	Satu user bisa punya banyak order
users → orders (booster)	0..1 : N	Booster mengerjakan banyak order; order bisa belum di-assign
joki_categories → joki_packages	1 : N	Kategori punya banyak paket/tier
joki_packages → orders	1 : N	Satu paket dipesan berkali-kali
orders → order_game_accounts	1 : 0..1	Dipisah agar data sensitif punya kebijakan akses & hapus sendiri
orders → payments	1 : N	Satu order bisa punya beberapa percobaan bayar (gagal lalu sukses)
orders → reviews	1 : 0..1	Satu ulasan per order
users → conversations	1 : 0..1	Satu thread chat per user
conversations → messages	1 : N	

Tidak ada relasi M:N murni, sehingga tidak perlu tabel pivot.

2.2 Normalisasi

- **1NF**: semua atribut atomik. Tidak ada kolom berulang atau multi-nilai (UID dan server dipisah kolom).
- **2NF**: semua tabel memakai PK tunggal (surrogate `id`), sehingga tidak ada dependensi parsial.
- **3NF**: tidak ada dependensi transitif. Data paket (nama, deskripsi) tidak diulang di `orders`, melainkan lewat FK. Kolom `payments`, `reviews`, dan `messages` dipisah dari `orders`.

- **Denormalisasi terkontrol**: `orders.price_snapshot` dan `est_hours_snapshot` sengaja menyalin nilai paket saat order dibuat agar riwayat keuangan tidak berubah. `conversations.*_unread` adalah counter untuk performa inbox.

3. DFD

Notasi: *external entity* = kotak, *proses* = lingkaran/rounded, *data store* = kotak berbingkai ganda (aproksimasi "open rectangle" di Mermaid).

3.1 Level 0: Context Diagram

```
flowchart LR
    U[User]
    A[Admin / Owner]
    PG[Payment Gateway]
    EM[Layanan Email]
    S(("0\nSistem Web Joki\nGenshin Impact"))

    U -->|"data registrasi, login, order, bukti bayar, chat, review"| S
    S -->|"status order, posisi antrian, invoice, leaderboard, notifikasi"| U
    A -->|"kelola katalog, status order, verifikasi bayar, balasan chat"| S
    S -->|"dashboard, daftar order, report keuangan, inbox chat"| A
    S -->|"permintaan transaksi, permintaan refund"| PG
    PG -->|"link/QR bayar, notifikasi status bayar (webhook)"| S
    S -->|"email verifikasi & reset password"| EM
```

3.2 Level 1

```
flowchart TB
    U[User]
    A[Admin / Owner]
    PG[Payment Gateway]

    P1(("1.0\nAutentikasi"))
    P2(("2.0\nManajemen Order"))
    P3(("3.0\nPembayaran"))
    P4(("4.0\nAntrian"))
    P5(("5.0\nChat"))
    P6(("6.0\nLeaderboard"))
    P7(("7.0\nReport Keuangan"))
    P8(("8.0\nManajemen Konten"))

    D1[["D1 users"]]
    D2[["D2 joki_categories & packages"]]
```

```
    D3[["D3 orders & order_game_accounts"]]
    D4[["D4 payments"]]
    D5[["D5 conversations & messages"]]
    D6[["D6 reviews"]]
    D7[["D7 activity_logs"]]

    U -->|"registrasi/login"| P1
    P1 -->|"token sesi"| U
    P1 <--> D1
    P1 --> D7

    U -->|"pilih paket, data akun, catatan"| P2
    P2 -->|"detail order, invoice"| U
    A -->|"ubah status, assign booster, batal"| P2
    P2 <--> D3
    P2 -->|"baca paket"| D2
    D2 --> P2
    P2 --> D7
    U -->|"review"| P2
    P2 --> D6

    U -->|"pilih metode, bukti transfer"| P3
    P3 -->|"status pembayaran"| U
    A -->|"verifikasi / refund"| P3
    P3 <-->|"transaksi & webhook"| PG
    P3 <--> D4
    P3 -->|"order jadi queued/refunded"| D3
    P3 --> D7

    D3 -->|"order queued"| P4
    P4 -->|"posisi & estimasi"| U
    A -->|"atur prioritas"| P4
    P4 -->|"update queue_priority"| D3

    U -->|"pesan"| P5
    P5 -->|"balasan"| U
    A -->|"balas pesan"| P5
    P5 -->|"inbox"| A
    P5 <--> D5

    D3 -->|"order completed"| P6
    D4 -->|"refund"| P6
    D1 -->|"username"| P6
    P6 -->|"ranking"| U
    A -->|"atur periode/reset"| P6
```

```
    D4 --> P7
    D3 --> P7
    D2 --> P7
    A -->|"pilih periode"| P7
    P7 -->|"report, grafik, PDF"| A

    A -->|"CRUD kategori/paket"| P8
    P8 <--> D2
    P8 -->|"ban/unban user"| D1
    D2 -->|"katalog"| P8
    P8 -->|"katalog"| U
```

3.3 Level 2: Proses 2.0 Manajemen Order

```
flowchart LR
    U[User]
    A[Admin / Owner]
    D2[["D2 packages"]]
    D3[["D3 orders & game_accounts"]]
    D7[["D7 activity_logs"]]

    P21(("2.1\nPilih Paket"))
    P22(("2.2\nInput Akun Game & Catatan"))
    P23(("2.3\nValidasi & Buat Order"))
    P24(("2.4\nUbah Status / Assign / Batal"))
    P25(("2.5\nTampilkan Detail & Invoice"))

    U -->|"pilih paket"| P21
    D2 -->|"harga, durasi"| P21
    P21 -->|"paket terpilih"| P22
    U -->|"UID, server, kredensial, catatan, lampiran"| P22
    P22 -->|"data terenkripsi"| P23
    P23 -->|"order status=pending_payment + snapshot harga"| D3
    P23 -->|"order_no"| U
    A -->|"status baru, booster, alasan batal"| P24
    D3 -->|"order saat ini"| P24
    P24 -->|"update order"| D3
    P24 --> D7
    D3 -->|"data order"| P25
    P25 -->|"detail & invoice PDF"| U
```

3.4 Level 2: Proses 3.0 Pembayaran

```
flowchart LR
    U[User]
    A[Admin / Owner]
    PG[Payment Gateway]
    D3[("D3 orders"]]
    D4[("D4 payments"]]
    D7[("D7 activity_logs"]]

    P31(("3.1\nBuat Transaksi"))
    P32(("3.2\nTerima Webhook Gateway"))
    P33(("3.3\nUpload & Verifikasi Bukti Manual"))
    P34(("3.4\nAktifkan Order ke Antrian"))
    P35(("3.5\nProses Refund"))

    U -->|"metode bayar"| P31
    D3 -->|"total tagihan"| P31
    P31 -->|"request transaksi"| PG
    PG -->|"redirect URL / QR"| P31
    P31 -->|"payment pending"| D4
    P31 -->|"link bayar"| U
    PG -->|"notifikasi status + signature"| P32
    P32 -->|"status paid/failed"| D4
    P32 -->|"payment paid"| P34
    U -->|"foto bukti transfer"| P33
    P33 -->|"bukti, status pending"| D4
    A -->|"setuju / tolak"| P33
    P33 -->|"payment paid/failed"| D4
    P33 -->|"payment paid"| P34
    P34 -->|"status=queued, paid_at"| D3
    P34 -->|"notifikasi"| U
    A -->|"refund order"| P35
    P35 -->|"request refund"| PG
    P35 -->|"payment refunded"| D4
    P35 -->|"status=cancelled"| D3
    P35 --> D7
```

3.5 Level 2: Proses 7.0 Report Keuangan

```
flowchart LR
    A[Admin / Owner]
    D2[("D2 packages & categories"]]
    D3[("D3 orders"]]
    D4[("D4 payments"]]
```

```
    P71(("7.1\nTentukan Periode Mingguan"))
    P72(("7.2\nAgregasi Pendapatan"))
    P73(("7.3\nBreakdown per Jenis Joki"))
    P74(("7.4\nHitung Sukses / Gagal / Refund"))
    P75(("7.5\nSusun Grafik Tren"))
    P76(("7.6\nExport PDF/Excel"))

    A -->|"minggu / rentang tanggal"| P71
    P71 -->|"start_date, end_date"| P72
    D4 -->|"payments paid"| P72
    D4 -->|"status payments"| P74
    D3 -->|"order + package_id"| P73
    D2 -->|"nama kategori"| P73
    P72 -->|"total per minggu"| P75
    P72 --> P73
    P73 -->|"revenue per kategori"| P76
    P74 -->|"jumlah transaksi"| P76
    P75 -->|"data grafik"| P76
    P76 -->|"dashboard + file PDF/Excel"| A
```

4. Skema SQL (PostgreSQL)

```
CREATE TYPE user_role      AS ENUM ('user','admin','booster');
CREATE TYPE user_status    AS ENUM ('active','banned');
CREATE TYPE order_status   AS ENUM
('pending_payment','queued','in_progress','completed','cancelled');
CREATE TYPE payment_method AS ENUM ('gateway','manual');
CREATE TYPE payment_status AS ENUM ('pending','paid','failed','refunded');
CREATE TYPE game_server    AS ENUM ('asia','america','europe','tw_hk_mo');

CREATE TABLE users (
  id BIGSERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(255) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  avatar_url VARCHAR(500),
  role user_role NOT NULL DEFAULT 'user',
  status user_status NOT NULL DEFAULT 'active',
  show_on_leaderboard BOOLEAN NOT NULL DEFAULT TRUE,
  email_verified_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_users_role_status ON users(role, status);
```

```
-- token verifikasi email & reset password: pakai tabel bawaan framework
-- (mis. password_reset_tokens) dengan token di-hash + expires_at.

CREATE TABLE joki_categories (
  id BIGSERIAL PRIMARY KEY,
  name VARCHAR(100) NOT NULL UNIQUE,
  slug VARCHAR(120) NOT NULL UNIQUE,
  description TEXT,
  is_active BOOLEAN NOT NULL DEFAULT TRUE
);

CREATE TABLE joki_packages (
  id BIGSERIAL PRIMARY KEY,
  category_id BIGINT NOT NULL REFERENCES joki_categories(id) ON DELETE
RESTRICT,
  name VARCHAR(150) NOT NULL,
  description TEXT,
  price NUMERIC(12,2) NOT NULL CHECK (price >= 0),
  estimated_hours INT NOT NULL CHECK (estimated_hours > 0),
  is_active BOOLEAN NOT NULL DEFAULT TRUE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_packages_category_active ON joki_packages(category_id,
is_active);

CREATE TABLE orders (
  id BIGSERIAL PRIMARY KEY,
  order_no VARCHAR(30) NOT NULL UNIQUE,            -- juga dipakai sebagai
nomor invoice
  user_id BIGINT NOT NULL REFERENCES users(id) ON DELETE RESTRICT,
  package_id BIGINT NOT NULL REFERENCES joki_packages(id) ON DELETE RESTRICT,
  booster_id BIGINT REFERENCES users(id) ON DELETE SET NULL,
  status order_status NOT NULL DEFAULT 'pending_payment',
  price_snapshot NUMERIC(12,2) NOT NULL,
  total_amount NUMERIC(12,2) NOT NULL,
  est_hours_snapshot INT NOT NULL,
  queue_priority INT NOT NULL DEFAULT 0,           -- makin besar makin
didahulukan
  customer_notes TEXT,
  cancel_reason TEXT,
  paid_at TIMESTAMPTZ,
  started_at TIMESTAMPTZ,
  completed_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
```

```
CREATE INDEX idx_orders_user_created ON orders(user_id, created_at DESC);
CREATE INDEX idx_orders_status ON orders(status);
CREATE INDEX idx_orders_booster_status ON orders(booster_id, status);
-- partial index khusus antrian:
CREATE INDEX idx_orders_queue ON orders(queue_priority DESC, paid_at ASC)
WHERE status = 'queued';
CREATE INDEX idx_orders_completed_at ON orders(completed_at) WHERE status =
'completed';

CREATE TABLE order_game_accounts (
  order_id BIGINT PRIMARY KEY REFERENCES orders(id) ON DELETE CASCADE,
  uid VARCHAR(20) NOT NULL,
  server game_server NOT NULL,
  credential_encrypted BYTEA NOT NULL,            -- AES-256-GCM (nonce +
ciphertext + tag)
  attachment_url VARCHAR(500),
  purge_at TIMESTAMPTZ                            -- diisi saat order
completed (+7 hari), dihapus oleh job terjadwal
);
CREATE INDEX idx_oga_purge ON order_game_accounts(purge_at) WHERE purge_at IS
NOT NULL;

CREATE TABLE payments (
  id BIGSERIAL PRIMARY KEY,
  order_id BIGINT NOT NULL REFERENCES orders(id) ON DELETE RESTRICT,
  method payment_method NOT NULL,
  gateway_ref VARCHAR(100) UNIQUE,                -- ID transaksi
Midtrans/Xendit (idempotensi webhook)
  amount NUMERIC(12,2) NOT NULL CHECK (amount > 0),
  status payment_status NOT NULL DEFAULT 'pending',
  proof_url VARCHAR(500),
  refunded_amount NUMERIC(12,2) NOT NULL DEFAULT 0,
  verified_by BIGINT REFERENCES users(id) ON DELETE SET NULL,
  paid_at TIMESTAMPTZ,
  refunded_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_payments_order ON payments(order_id);
CREATE INDEX idx_payments_status_paid_at ON payments(status, paid_at);

CREATE TABLE reviews (
  id BIGSERIAL PRIMARY KEY,
  order_id BIGINT NOT NULL UNIQUE REFERENCES orders(id) ON DELETE CASCADE,
  user_id BIGINT NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  rating SMALLINT NOT NULL CHECK (rating BETWEEN 1 AND 5),
  comment TEXT,
```

```
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE conversations (
  id BIGSERIAL PRIMARY KEY,
  user_id BIGINT NOT NULL UNIQUE REFERENCES users(id) ON DELETE CASCADE,
  last_message_at TIMESTAMPTZ,
  admin_unread INT NOT NULL DEFAULT 0,
  user_unread INT NOT NULL DEFAULT 0
);
CREATE INDEX idx_conv_last_msg ON conversations(last_message_at DESC);

CREATE TABLE messages (
  id BIGSERIAL PRIMARY KEY,
  conversation_id BIGINT NOT NULL REFERENCES conversations(id) ON DELETE
CASCADE,
  sender_id BIGINT NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  order_id BIGINT REFERENCES orders(id) ON DELETE SET NULL,
  body TEXT NOT NULL,
  attachment_url VARCHAR(500),
  read_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_messages_conv_created ON messages(conversation_id,
created_at DESC);

CREATE TABLE activity_logs (
  id BIGSERIAL PRIMARY KEY,
  actor_id BIGINT REFERENCES users(id) ON DELETE SET NULL,
  action VARCHAR(80) NOT NULL,                    -- mis.
order.status_changed, payment.refunded, user.banned
  entity_type VARCHAR(50),
  entity_id BIGINT,
  meta JSONB,
  ip INET,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX idx_logs_entity ON activity_logs(entity_type, entity_id);
CREATE INDEX idx_logs_created ON activity_logs(created_at DESC);

-- Leaderboard all-time (order selesai, dikurangi refund)
CREATE VIEW v_leaderboard_alltime AS
SELECT u.id AS user_id,
       CASE WHEN u.show_on_leaderboard THEN u.name ELSE 'Anonim' END AS
display_name,
       SUM(o.total_amount) AS total_spending,
```

```
       COUNT(*) AS total_orders,
       RANK() OVER (ORDER BY SUM(o.total_amount) DESC) AS rank
FROM orders o JOIN users u ON u.id = o.user_id
WHERE o.status = 'completed'
GROUP BY u.id, u.name, u.show_on_leaderboard;

-- Periode mingguan/bulanan: query yang sama + WHERE o.completed_at >=
date_trunc('week'|'month', now()).
-- Untuk trafik tinggi: cache hasil di Redis (TTL 5 menit) atau MATERIALIZED
VIEW.

-- Posisi antrian suatu order:
-- SELECT COUNT(*) + 1 FROM orders q, orders me
-- WHERE me.id = :id AND q.status = 'queued'
--   AND (q.queue_priority > me.queue_priority
--        OR (q.queue_priority = me.queue_priority AND q.paid_at <
me.paid_at));
-- Estimasi tunggu = SUM(est_hours_snapshot order di depan) / jumlah booster
aktif.

-- Report mingguan:
-- SELECT date_trunc('week', p.paid_at) AS minggu, c.name AS kategori,
--        SUM(p.amount - p.refunded_amount) AS pendapatan, COUNT(*) AS
transaksi
-- FROM payments p JOIN orders o ON o.id = p.order_id
-- JOIN joki_packages k ON k.id = o.package_id JOIN joki_categories c ON c.id
= k.category_id
-- WHERE p.status IN ('paid','refunded') GROUP BY 1, 2 ORDER BY 1;
```

Rekomendasi indexing tambahan: pantau query lambat (`pg_stat_statements`), lalu tambahkan index sesuai kebutuhan. Hindari index berlebihan di tabel yang sering ditulis (`messages` , `activity_logs`).

5. Rekomendasi Tech Stack

Lapisan	Rekomendasi utama	Alternatif	Alasan
Backend	**Laravel 11**	NestJS	Auth, verifikasi email, reset password, queue, scheduler, policy, rate limiter, dan paket PDF/Excel sudah tersedia. Cepat untuk 1–2 developer.
Frontend	**Inertia.js + Vue/React** (Tailwind CSS)	Next.js	Satu codebase dengan backend, lebih sedikit boilerplate API. Pakai Next.js jika ingin FE terpisah.
Database	**PostgreSQL 16**	MySQL 8	Partial index, JSONB, dan window function (leaderboard).
Cache/Queue/Session	**Redis**	–	Session, rate limit, cache leaderboard, queue job (email, webhook).
Realtime chat	**Laravel Reverb** (WebSocket)	Socket.io, Pusher	Reverb self-hosted dan gratis. Pusher lebih mudah tapi berbayar. Fallback: polling 5 detik.
Payment	**Midtrans Snap** (QRIS, VA, e-wallet)	Xendit	Dokumentasi bagus dan sandbox mudah. Wajib verifikasi signature webhook.
Storage	**Cloudflare R2** (S3-compatible)	S3, Cloudinary	Murah, tanpa biaya egress. Bukti bayar & foto profil memakai signed URL.

Lapisan	Rekomendasi utama	Alternatif	Alasan
Report	`barryvdh/laravel-dompdf` + `maatwebsite/excel`, grafik Chart.js	–	Export PDF/Excel dan grafik tren.
Hosting	**VPS** (Nginx + PHP-FPM + PostgreSQL + Redis), region Singapura/Indonesia	Railway, Fly.io	Biaya stabil dan kontrol penuh. Vercel hanya cocok jika memakai Next.js terpisah.
Backup & monitoring	`pg_dump` harian → R2 (retensi 14 hari), Sentry, UptimeRobot	–	Memenuhi NFR-09.
Keamanan	Cloudflare (WAF, DDoS), Let's Encrypt, argon2id	–	Memenuhi NFR-01 s/d NFR-04.

6. Sitemap

Publik & User

```
/                          Beranda (hero, kategori, testimoni)
/layanan                   Daftar kategori & paket
/layanan/{slug}            Detail kategori/paket
/leaderboard               Leaderboard
/auth/register | /login | /forgot-password | /reset-password | /verify-email
/akun                      (login)
 ├─ /profil                Edit profil & ganti password
 ├─ /pesanan               Riwayat order
 |   └─ /pesanan/{no}      Detail order: status, antrian, pembayaran, review,
unduh invoice
 ├─ /checkout/{package}    Form akun game, catatan, lampiran, disclaimer
 ├─ /bayar/{no}            Pilih metode & upload bukti
 └─ /chat                  Chat dengan admin
```

Admin (`/admin` , wajib role `admin`)

```
/admin/dashboard
/admin/layanan            CRUD kategori & paket
/admin/pesanan            Daftar order + filter status → detail (ubah status,
assign booster, batal, refund)
/admin/antrian            Urutkan / prioritaskan antrian
/admin/pembayaran         Verifikasi bukti manual + riwayat
/admin/user               Daftar user → detail (ban/unban, reset)
/admin/chat               Inbox
/admin/report             Report keuangan mingguan + grafik + export
/admin/leaderboard        Atur periode / reset (opsional)
/admin/log                Log aktivitas (opsional)
```

Booster (opsional) memakai `/booster/tugas`: daftar order yang di-assign dan tombol mulai/selesai.

7. User Flow

7.1 Order → Payment → Antrian → Selesai → Review

```
flowchart TD
    A([Mulai]) --> B[Login / Register]
    B --> C[Pilih kategori & paket]
    C --> D[Isi UID, server, kredensial, catatan, lampiran]
    D --> E[Setuju disclaimer & Checkout]
    E --> F[Order dibuat: Menunggu Pembayaran]
    F --> G{Metode bayar}
    G -->|Gateway| H[Redirect ke Midtrans]
    G -->|Manual| I[Upload bukti transfer]
    H --> J{Webhook status}
    I --> K{Admin verifikasi}
    J -->|paid| L[Order: Menunggu Antrian]
    K -->|valid| L
    J -->|gagal / kedaluwarsa| M[Pembayaran gagal]
    K -->|ditolak| M
    M --> N{Coba lagi?}
    N -->|ya| G
    N -->|tidak / 24 jam| X[Order Dibatalkan]
    L --> O[User melihat posisi antrian & estimasi]
    O --> P[Admin assign booster]
    P --> Q[Order: Dikerjakan]
    Q --> R[Booster selesai + bukti]
    R --> S[Order: Selesai, kredensial dijadwalkan dihapus]
    S --> T[User unduh invoice, masuk leaderboard]
```

```
    T --> U[User memberi review]
    U --> Z([Selesai])
    L -.->|admin refund| W[Refund → Order Dibatalkan]
    Q -.->|admin refund| W
```

7.2 Alur Chat

```
flowchart TD
    A([User buka /chat]) --> B[Sistem ambil / buat conversation]
    B --> C[Muat riwayat pesan]
    C --> D[User kirim pesan + lampiran opsional]
    D --> E[Server validasi + simpan ke messages]
    E --> F[Broadcast WebSocket ke channel admin]
    F --> G{Admin online?}
    G -->|ya| H[Inbox admin update + badge unread]
    G -->|tidak| I[Tersimpan; notifikasi email/badge saat admin login]
    H --> J[Admin membalas]
    I --> J
    J --> K[Simpan pesan + broadcast ke user]
    K --> L{User online?}
    L -->|ya| M[Pesan tampil, read_at diisi]
    L -->|tidak| N[user_unread bertambah, notifikasi email]
    M --> D
    N --> D
```

8. Asumsi & Catatan Penting

#	Topik	Keputusan	Alasan
1	Multi-jenis dalam 1 order	**Tidak**, 1 order = 1 paket	Antrian, harga, dan review lebih sederhana. Bisa dikembangkan jadi `order_items` nanti.
2	Booster terpisah dari admin	**Ya**, lewat `users.role='booster'` dan `orders.booster_id`	Owner bisa menambah pengerja tanpa ubah skema. Jika awalnya hanya owner, `booster_id` boleh dikosongkan.
3	Pembayaran pending	Kedaluwarsa otomatis 24 jam, order → `cancelled` (job terjadwal)	Mencegah antrian dan data menumpuk.

#	Topik	Keputusan	Alasan
4	Refund	Penuh atau sebagian, tercatat di `payments.refunded_amount`; leaderboard dan report memakai nilai bersih	Keuangan konsisten.
5	Leaderboard	Hanya order `completed`; opsi anonim	Adil dan menjaga privasi.
6	Pengubahan status order	Hanya transisi valid: `pending_payment→queued→in_progress→completed`, plus `→cancelled`; semua dicatat di `activity_logs`	Mencegah lompat status dan memudahkan audit.
7	Estimasi antrian	Total jam order di depan ÷ jumlah booster aktif; ditampilkan sebagai perkiraan	Sederhana dan cukup akurat.
8	Webhook	Idempotent (cek `gateway_ref` unik + signature)	Mencegah pembayaran ganda tercatat.
9	Chat	1 thread per user, pesan bisa merujuk ke order tertentu	Inbox admin sederhana.
10	Penyimpanan kredensial game	Terenkripsi di aplikasi, kunci di env/KMS, akses hanya booster yang di-assign, akses dicatat di log, dihapus otomatis	Meminimalkan risiko kebocoran.
11	Rencana pengembangan (fase 2)	Review, export Excel, filter leaderboard periodik, notifikasi push, kupon/diskon, tabel `order_items` bila diperlukan	Fokus MVP dulu.

Urutan pengerjaan yang disarankan: (1) Auth + katalog → (2) Order + payment gateway → (3) Antrian + admin order → (4) Chat → (5) Leaderboard + report → (6) Review + log + polish.